

SHREY GROVER

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& Brain Sciences, Boston University, Boston, MA – 02215

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EMPLOYMENT & EDUCATION

2024 – Present	Boston University Postdoctoral Associate (PI: Robert MG Reinhart, PhD)
2017 – 2024	Boston University PhD in Psychology (Brain, Behavior & Cognition) (PI: Robert MG Reinhart, PhD)
2015 – 2017	Indian Institute of Science Research Assistant, Centre for Neuroscience (PI: Devarajan Sridharan, PhD)
2011 – 2015	Birla Institute of Technology & Science, Pilani BE (Hons) in Mechanical Engineering

PEER-REVIEWED PUBLICATIONS

1. Grover S, Wen W, Reinhart RMG (In press).
Neuromodulation for restoring and amplifying brain function.
[Nature Neuroscience](#)
2. Lin C*, Wen W*, Cheng P, Schallies S, Grover S, Reinhart RMG (2025).
Salient experiences enhance mundane memories through graded prioritization.
[Science Advances](#)
3. Rogers D*, O'Brien WJ*, Gao Y, Zimmermann B, Grover S, et al. (2025).
Co-localized optode-electrode design for multimodal electroencephalography and high-density functional near infrared spectroscopy.
[Neurophotonics](#)
4. Wen W, Grover S, Hazel D, Berning P, Baumgardt F, Viswanathan V, Tween O, Reinhart RMG (2024).
Beta frequency neural variability reveals age-related dissociations in human working memory maintenance and deletion.
[PLoS Biology](#)
5. Sengupta A*, Banerjee S*, Ganesh S, Grover S, Sridharan D (2024).
The right posterior parietal cortex controls spatial reorienting of attentional choice bias.
[Nature Communications](#)
6. Molho W, Raymond N, Reinhart RMG, Trotti R, Grover S, Keshavan M, Lizano P (2023).
Lesion network guided delta frequency neuromodulation improves cognition in patients with psychosis

spectrum disorders: A pilot study.

[Asian Journal of Psychiatry](#)

7. Raymond N, Reinhart RMG, Trotti R, Parker D, Grover S, et al. (2023).
A pilot study to investigate the efficacy and tolerability of lesion network guided transcranial electrical stimulation in outpatients with psychosis spectrum illness.
[Asian Journal of Psychiatry](#)
8. Grover S, Fayzullina R, Bullard BM, Levina V, Reinhart RMG (2023).
A meta-analysis suggests that tACS improves cognition in healthy, aging, and psychiatric populations.
[Science Translational Medicine](#)
9. Cheng P, Grover S, Wen W, Sankaranarayanan S, Fragetta J, Davies S, Soto D, Reinhart RMG (2022).
Dissociable rhythmic mechanisms enhance memory for conscious and nonconscious perceptual contents.
[Proceedings of the National Academy of Sciences U. S. A.](#)
10. Grover S, Wen W, Viswanathan V, Gill CT, Reinhart RMG (2022).
Long-lasting, dissociable improvements in working memory and long-term memory in older adults with repetitive neuromodulation.
[Nature Neuroscience](#)

Commentary: Rogers, J (2022). A frequency location to remember.
Nature Reviews Neuroscience
11. Grover S, Nguyen JA, Viswanathan V, Reinhart RMG (2021).
High-definition neuromodulation improves obsessive-compulsive behavior.
[Nature Medicine](#)

Commentary: Figuee M, Mayberg H (2021). The future of personalized brain stimulation.
Nature Medicine

Commentary: Frohlich F, Riddle J, Abramowitz JS (2021). Transcranial alternating current stimulation for the treatment of obsessive-compulsive disorder?
Brain Stimulation
12. Grover S, Nguyen JA, Viswanathan V, Reinhart RMG (2021).
Synchronizing brain rhythms to improve cognition.
[Annual Review of Medicine](#)
13. Banerjee S*, Grover S*, Ganesh S, Sridharan D (2019).
Sensory and decisional components of endogenous attention are dissociable.
[Journal of Neurophysiology](#)
14. Banerjee S, Grover S, Sridharan D (2017).
Unraveling causal mechanisms of top-down and bottom-up visuospatial attention with non-invasive brain stimulation.
[Journal of The Indian Institute of Science](#)

(* equal contribution)

CHAPTERS & COMMENTARIES

1. Grover S, Reinhart RMG (In press)
Neuromodulation of memory function in aging: Mechanisms, evidence, and clinical prospects
In Clark S et al. (Ed 3). *Neural Repair and Rehabilitation*.
Cambridge University Press.
2. Grover S, Keshavan MS, Lizano PL, Reinhart RMG (2021).
Proximate markers of cognitive dysfunction in schizophrenia.
Schizophrenia Research
3. Grover S, Reinhart RMG (2020).
Modulating anterior midcingulate cortex using theta burst stimulation.
Biological Psychiatry: Cognitive Neuroscience & Neuroimaging
4. Grover S, Reinhart RMG (2019).
Combining transcranial stimulation and electrophysiology to understand the memory representations that guide attention.
In Pollman S. (Ed). *Spatial Learning and Attention Guidance*.
Neuromethods, Springer Nature. New York, NY. pp. 1-29.

UNDER REVIEW

1. Grover S, Qiu L, Scangos K, Halpern CH, Reinhart RMG.
Toward a convergent biomarker and treatment for obsessive-compulsive disorder.
2. Wen W, Grover S, Wang, C, Fang Yu, Lin C, Hazel D, Schallies S, Yu H, Reinhart RMG.
A striatal β rhythm accelerates instruction-to-action in humans.
3. Lin C, Cao R, Wen W, Fang Y, Grover S, Zhu J, Hoang J, Zeng W, Hayashi M, Huang Y, Reinhart RMG.
Theta rhythm sets the pace of temporal computation in human striatum and hippocampus.

IN PREPARATION

1. Grover S, Zhou Y, Wen W, Cao F, Hemavathy SGMS, Arora M, Wang J, Han X, Reinhart RMG.
Bidirectional control of striatal pathways: Cross-species rhythmic motifs for noninvasive neuromodulation.
2. Grover S*, Halder S*, Pek T, Lin C, Wang J, Reinhart RMG.
Shared and distinct contributions of the basal ganglia during control of attention and working memory.
3. Grover S, Cao F, Cheng P, Reinhart RMG.
Interhemispheric synchrony causally shapes rhythmic attentional sampling.
4. Wen W, Grover S, Hazel D, Reinhart RMG.
Bidirectional control of impulsivity by integrating or segregating rhythmic brain circuits.

CONFERENCE PROCEEDINGS

1. Raymond N, Trotti R, Stiltner B, Molho W, Ying D, Grover S, et al. (2026). Differential effects of beta-gamma and direct current tES in individuals with mania and psychosis. *American Psychopathological Association. Boston, U. S. A.*,
2. Grover S, Cao F, Cheng P, Reinhart RMG (2026). Interhemispheric synchrony causally shapes rhythmic attentional sampling. *Vision Science Society. St. Pete's Beach, U. S. A.*
3. Wen W, Grover S, Hazel D, Berning P, Baumgardt F, Viswanathan V, Tween O, Reinhart RMG (2024). Beta band neural variability marks age-related dissociations in working memory maintenance and deletion. *Gordon Research Conference. New Hampshire, U. S. A.*
4. Rogers D, O'Brien WJ, Gao Y, Zimmerman B, Grover S, et al. (2023). Co-localized optodes and electrodes allow for simultaneous HD-fNIRS-EEG measurements. *Organization for Human Brain Mapping. Montréal, Canada.*
5. Cheng P, Sankaranarayanan S, Davies S, Fragetta J, Grover S, Wen W, Soto D, Reinhart RMG (2022). β and α entrainment preferentially enhance memory for conscious and nonconscious information. *Society for Neuroscience. San Diego, U. S. A.*
6. Grover S, Nguyen JA, Viswanathan V, Reinhart RMG (2021). Orbitofrontal beta-gamma rhythms influence reward processing and obsessive-compulsive behaviors. *Society for Neuroscience. (Virtual).*
7. Wen W, Grover S, Viswanathan V, Gill CT, Reinhart RMG (2021). Enduring memory improvements in older adults using high-definition transcranial alternating current stimulation. *Society for Neuroscience. (Virtual).*
8. Fayzullina R, Grover S, Bullard BM, Levina V, Reinhart RMG (2021). Does transcranial alternating current stimulation modulate cognition? A systematic review and meta-analysis. *Society for Neuroscience. (Virtual).*
9. Bullard BM, Levina V, Grover S, Reinhart RMG (2020). Effects of transcranial alternating current stimulation on visual cognition: A systematic review and meta-analysis. *Vision Science Society. (Virtual).*
10. Reinhart RMG, Grover S, Wang CX, Nguyen JA (2019). Improving working memory in older adults by synchronizing cortical interactions with alternating current. *3rd International Brain Stimulation Conference. Vancouver, Canada.*
11. Sreenivasan V, Grover S, Sridharan D (2017). Structural connectivity correlates of behavioural asymmetries in attention measured with diffusion MRI

tractography.

4th Annual Conference on Cognitive Science. Hyderabad, India.

12. Grover S, Gurusamy G, Purokayastha S, Ramesh RG, Javali M, Rao NP, Sridharan D (2017). Behavioral indices for quantifying attention deficits in Alzheimer's disease. *Alzheimer's & Dementia: The Journal of the Alzheimer's Association. London, United Kingdom.*
13. Sridharan D, Banerjee S, Grover S, Nabeel A, Abhimanyu P, Purokayastha S, Sriram R, Sahoo B, Sreenivasan V (2016). Selective attention: Model-based approaches for identifying network mechanisms. (equal contribution). *The 39th Annual Meeting of the Japan Neuroscience Society. Tokyo, Japan.*
14. Grover S, Banerjee S, Rajagopalan J, Gupta S, Abhimanyu P, Sridharan D (2015). Investigating the role of rhythmic stimulation of the pre-frontal and parietal cortex in selective attention. *International Conference on Neurodegenerative Diseases: From Pathogenesis to Therapy. Bengaluru, India.*
15. Grover S, Banerjee S, Rajagopalan J, Gupta S, Abhimanyu P, Sridharan D (2015). Investigating the role of rhythmic stimulation of the pre-frontal and parietal cortex in selective attention. *1st Symposium on Non-Invasive Brain Stimulation in Psychiatry. Bengaluru, India.*

HONORS

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| 2024 | Kavita Jain Dissertation Award
Psychological & Brain Sciences, Boston University |
| 2017 | Dean's Fellowship
Psychological & Brain Sciences, Boston University |

TALKS

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| Feb 2026 | Striatal modulation of reward-guided behaviors
<i>Han Lab, Department of Biomedical Engineering, Boston University, Boston, MA</i> |
| Aug 2025 | Cell-type specific temporal interference stimulation of the human striatum
<i>Han Lab, Department of Biomedical Engineering, Boston University, Boston, MA</i> |
| Jan 2025 | Memory enhancements in aging with transcranial alternating current stimulation
<i>Department of Geriatric Psychiatry, Tufts Medical Center, Somerville, MA</i> |
| Aug 2024 | Long- and short-term memory modulation in aging with noninvasive neuromodulation
<i>Center for Translational Cognitive Neuroscience, Boston Medical Center, Boston, MA</i> |
| Oct 2023 | Brain rhythms and their modulation: Causal investigations and translational implications
<i>Cognition Lab, Indian Institute of Science, Bengaluru, India</i> |
| Apr 2023 | High-frequency neuromodulation improves obsessive-compulsive behaviors
<i>Symposium: Transcranial Electrical Stimulation and Its Applications in Psychiatry Society for Biological Psychiatry, San Diego, CA</i> |

Mar 2022	Does transcranial alternating current stimulation (tACS) improve cognitive function? A meta-analytic investigation. <i>Department of Psychological & Brain Sciences, Boston University</i>
Apr 2021	High-frequency neuromodulation improves obsessive-compulsive behaviors <i>COBRE Center for Neuromodulation, Brown University</i>
Mar 2021	High-frequency neuromodulation improves obsessive-compulsive behaviors <i>Department of Psychological & Brain Sciences, Boston University</i>
Dec 2018	<i>Oscillatory Dynamics of Sequence Learning</i> Department of Psychological & Brain Sciences, Boston University

TEACHING EXPERIENCE

PS/NE 329	Experimental Psychology: Cognitive Neuroscience Psychological & Brain Sciences, Boston University	Guest Lecturer (Feb 2025, Feb 2023, Mar 2022)
PS/NE 333	Drugs and Behavior Psychological & Brain Sciences, Boston University	Teaching Assistant (Fall 2017)

RESEARCH MENTORSHIP

Current Research Assistants

2026 – Present	Connie Lin (Boston University)	Research Assistant at Boston University
2026 – Present	Sophia Caires (Boston University)	Research Assistant at Boston University
2025 – Present	Tony Pek (Boston University)	Research Assistant at Boston University
2025 – Present	Sreenisha Gana Sai Murali Hemavathy (Boston University)	Research Technician at Boston University
2025 – Present	Myra Arora (Boston University)	UROP Scholar at Boston University
2023 – Present	Sara Bissell (Brandeis University)	Research Assistant at Boston University

Former Research Assistants

2025 – 2026	Junzheng Wang (Undergraduate student at BU)	Next position: RA at Massachusetts General Hospital, Boston, MA
2025 – 2026	Sarah Lehner (Master's student at BU)	Next Position: Psychometrist at LifeStance Health, Boston, MA
2023 – 2025	Feiyang Cao (Master's student at BU)	Next position: PhD student at University of Melbourne, Melbourne, Australia

2019 – 2020	Weijia Zhang (Undergraduate student at BU)	Next position: Masters student at Columbia University
2019 – 2020	Chenlingxi Xu (Master's student at BU)	Next position: PhD student at Notre Dame University
2017 – 2020	Charlotte Wang (Undergraduate student at BU)	Next position: RA at Massachusetts Institute of Technology

AD-HOC REVIEWING SERVICE

Asian Journal of Psychiatry
 Attention, Perception & Psychophysics
 Biological Psychiatry
 Biological Psychology
 Cell Reports
 Cerebral Cortex
 Cortex
 Current Biology
 Journal of Cognitive Neuroscience
 Journal of NeuroEngineering and Rehabilitation
 Journal of Neuroscience
 Journal of Physiology
 Molecular Psychiatry
 Nature Biotechnology
 Nature Communication
 Nature Medicine
 Nature Mental Health
 Nature Neuroscience
 Neurobiology of Aging
 NeuroImage
 Neuron
 Neuropsychologia
 Patterns
 PLoS Biology
 PLoS One
 PNAS
 Quarterly Journal of Experimental Psychology
 Schizophrenia Research

PATENTS

Krishnan JN, Grover S, Agarwal A, Deshpande A (2024).
 Continuous flow microfluidic device for rapid heavy metal ion detection in water samples.
India Patent #526,000.

SELECT MEDIA COVERAGE

05/06/2023	Does brain stimulation boost memory and focus? Mega study tries to settle debate. Nature
08/23/2022	Using electricity, researchers find surprising memory results. Psychology Today
08/22/2022	Brain stimulation leads to long-lasting improvements in memory. Nature
08/22/2022	Electrical brain stimulation improves memory, new study shows. The Wall Street Journal
08/22/2022	Brain stimulation boosts memory for a month. BBC
08/22/2022	Brain stimulation can improve the memory of older people. MIT Technology Review
08/22/2022	Electrical brain stimulation improves memory in elderly, research find. Financial Times of London
08/22/2022	Electrically zapping specific brain regions can boost memory. The Scientist
08/22/2022	Can you improve memory? Brain stimulation could make a moderate to large difference, study shows. USA Today
03/08/2021	Stimulation sensation: The non-invasive technologies shaking up neuroscience. Technology Networks
02/24/2021	A better way to zap our brains. The New York Times
01/19/2021	Electrical brain stimulation may alleviate obsessive-compulsive behaviors. Scientific American
01/18/2021	Researches show efficacy of personalized brain stimulation in psychiatric treatments. NPR